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Weather-Label-Free Salient Object Detection with an Inert Spatial Router

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Abstract

Salient object detection degrades under adverse weather. We ask whether spatial token-level mixture-of-experts (MoE) routing, used without weather labels, helps. On WXSOD the model reaches MAE 0.0192 on 1,500 synthetic and 0.0168 on 554 real test images, competitive with weather-supervised methods. A full ablation shows the mixture contributes nothing measurable: the no-MoE, one-scale and no-load-balance variants each sit within the seed-noise band of the reference and pass a TOST equivalence test at a margin of 0.001 MAE, while the dense-expert variant does not. The arms differ from the reference in expert count as well as the named field and share its training seed, so this is not an independent-seed comparison. Entropy sits at 99.8–99.97% of its ceiling for the reported model, and at or above 99.2% in every functional arm except the single-expert arm A3 (97.7% at 1/8), yet the hard assignment is strongly skewed: a near-uniform gate coexists with a consistent ranking. The numbers are not attributable to the mixture; the backbone and decoder were not ablated, so we do not attribute them there either. Section 5 asks why, and presents ranked hypotheses: the leading one is that the experts are near-identical at initialisation, leaving the gate nothing to separate, which makes the gate a static selector. The decisive measurements (initial-versus-trained top-2 assignment agreement, and the input-dependence of the routing) have not been made. We state the boundary of what this supports: one backbone, one dataset and one configuration. The equivalence is established; the mechanism is presented as hypothesis rather than fact.

Keywords salient object detection · mixture-of-experts · adverse weather · sparse routing · cross-attention decoder

1 Introduction

SOD identifies visually prominent regions as pixel-wise masks [1,2], but performance degrades under adverse weather. Two strategies address this: (1) weather-label-conditioned models (NIFM [3]), which require metadata at inference; (2) MoE for weather restoration (WM-MoE [4]), which targets pixel regression, not segmentation. Existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for multi-modal inputs rather than per-token routing for single-RGB SOD. We build such a combination (spatial token-level routing, independent routers at several pyramid scales, and routing-informed decoding) for single-RGB SOD without weather supervision, and the question this paper answers is whether it helps. Sparse MoE has been studied in dense prediction: MoE-SPNet routes at the scene level [5], token-level routing has been applied to medical segmentation (SegMoTE [6]), and Pavlitska et al. examine the behaviour of sparse MoE layers in CNN segmentation [7], and spatial MoE (SMoE) was introduced for dense regression on spatially-structured scientific data, such as weather and climate fields [8]. None of these addresses adverse-weather SOD with a single RGB input.

This paper's central finding is a null result. We designed and trained a spatial MoE that routes individual tokens to top-K experts at each of three pyramid scales, with per-token routing entropy fed to the decoder, and it does not measurably affect predictions. The architecture is competitive: it reaches MAE 0.0168 on the real test split without ever seeing a weather label. Yet eleven arms of one ablation family, plus the reported model and three earlier-recipe arms, and three inference-time interventions show that the mixture-of-experts, its routing and its entropy channel can each be removed, randomised or short-circuited with no effect larger than the seed-variance floor (Sections 4.4 and 4.5). We report the architecture and its numbers together with the evidence that they are not attributable to routing.

Our contributions are:

1) A token-level spatial router that assigns each token to top-K=2 experts from 8 per scale, using depthwise-convolutional context, noisy top-K gating [9], and true sparse dispatch.
2) Three independent routers at scales 1/4, 1/8, 1/16, each with a separate expert pool.
3) Routing entropy over the full expert distribution, normalized by $\log E$ for E=8, is injected as an explicit decoder feature via learnable additive fusion.
4) An ablation and inference-time intervention study (11 trained arms, 3 interventions) showing that this routing mechanism is inert.
5) A candidate explanation of *why* it is inert (Section 5), presented as ranked hypotheses: noise-dominated selection at initialisation; auxiliary and Z-loss terms that do not reward decisiveness; and experts that are near-identical at initialisation, giving the gate nothing to separate and leaving a static selector. Each is stated with the evidence bearing on it and the measurement that would discriminate it. We keep the equivalence result separate from the mechanism and do not claim the mechanism is settled.

The rest of the paper is organised as follows. Section 2 positions the work against adverse-weather SOD and mixture-of-experts methods. Section 3 describes the architecture: the multi-scale backbone, the spatial router, sparse dispatch, the entropy-fused decoder and the training objective. Section 4 reports the results: the headline numbers, a comparison against published methods, and a controlled ablation suite.

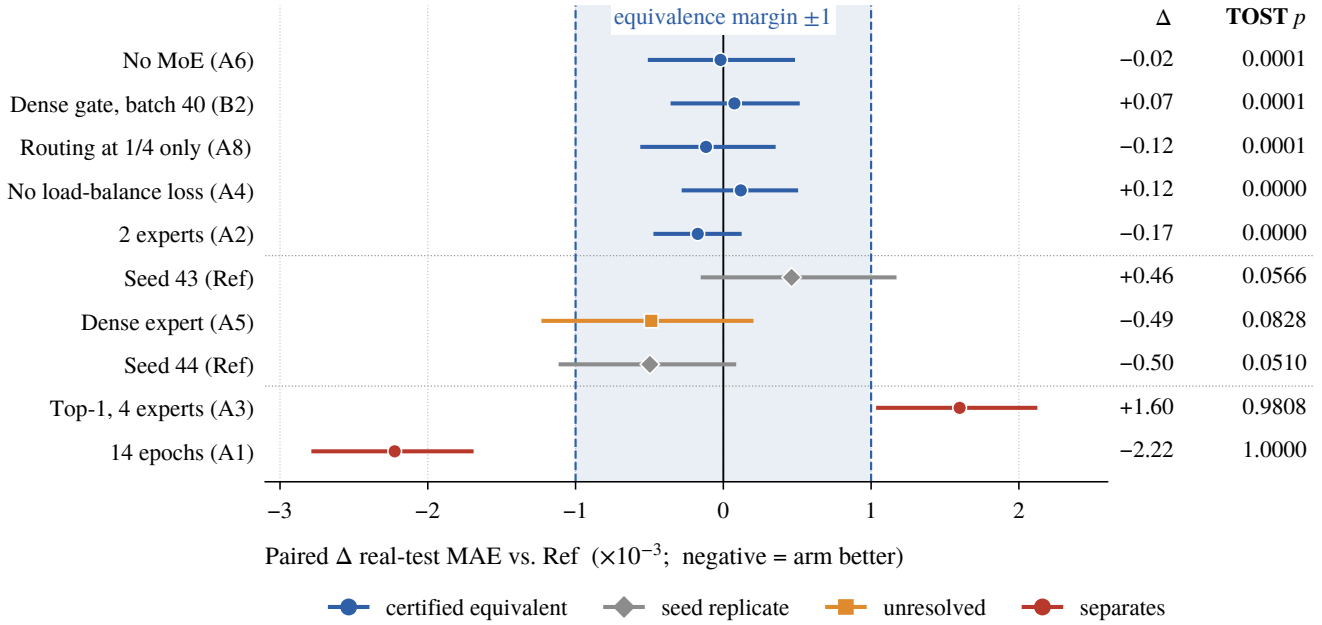


Fig. 1 Removing or reducing the mixture leaves MAE inside the equivalence band. Paired difference in real-test MAE between each ablation arm and the reference ($\times 10^{-3}$; 95% bootstrap interval over the 554 test images; negative means the arm is better). The shaded band is the ±0.001 TOST margin. Of the plotted arms, A6, A8, A4, A2 and B2 are certified equivalent, A3 and A1 fall outside the band, and A5 and the two seed replicates are unresolved. Arms share the reference's seed and differ from it in expert count as well as the named field (Section 4.4); the diverged run B3 (Δ = +0.0352) is off the axis. Full numbers: Table 6.

Section 5 analyses why the ablation behaves as it does. Section 6 states what the evidence supports and what it does not. Data and code availability are stated at the end of the paper.

2 Related Work

2.1 Adverse-Weather Salient Object Detection

Salient object detection under adverse weather is a comparatively recent benchmark setting. Robustness to adverse conditions is itself an established evaluation axis [10, 11]; WXSOD [12] supplies the benchmark used here and its own weather-aware baseline, WFANet, which trains a weather classifier over nine categories and conditions feature fusion on its output. NIFM [3] fuses noise indicators derived from explicit weather-type one-hot vectors. Both strategies therefore require weather supervision at training time, and the class-conditioned variant requires it at inference as well. Our model uses no weather labels at any point; this is the property that makes the comparison in Section 4.3 meaningful.

2.2 Mixture-of-Experts for Adverse-Weather Restoration

Mixture-of-experts has instead been applied to adverse-weather restoration [13, 14], notably WM-MoE [4], which routes with weather-aware gating and targets pixel regression rather than segmentation. The distinction matters: restoration is a per-pixel reconstruction task in which a small residual error is the objective, whereas SOD is a classification-like segmentation task in which the decision is largely made by the backbone's features. Our results are consistent with routing capacity not being the binding constraint here, though that is an observation from one backbone and one dataset rather than a general claim about segmentation.

2.3 Mixture-of-Experts in Dense Prediction

Within dense prediction, MoE-SPNet routes at the scene level [5]; SegMoTE applies token-level routing to medical image segmentation [6]; and Pavlitska et al. study the behaviour of sparse MoE layers in CNN semantic segmentation [7, 15]. In multi-task and multi-modal settings these routers are task- or modality-conditioned [16–18]; within SOD, existing MoE methods explore modality-, scale- or task-level routing for multi-modal inputs rather than per-token routing for single-RGB imagery.

2.4 Token-Level Routing and Routing Entropy

Sparse MoE goes back to adaptive mixtures of local experts [19]; token-level top-K routing was established by V-MoE [20] for classification, building on sparsely-gated MoE [9] and the GShard/Switch/ST-MoE line [21–23], with Mixtral scaling it further [24], where routing entropy has also served for uncertainty estimation; we instead inject it as a decoder feature for dense prediction. Spatial MoE (SMoE) was introduced for dense regression on spatially-structured scientific data [8]. We apply this combination to single-RGB adverse-weather SOD, and the question of whether it helps is one this paper answers in the negative.

3 Method

Fig. 2 illustrates the full architecture. A shared PVTv2-B4 backbone extracts multi-scale features; three independent SpatialMoE layers process tokens at each scale; a cross-attention decoder fuses the results and produces dense predictions. We detail each component below.

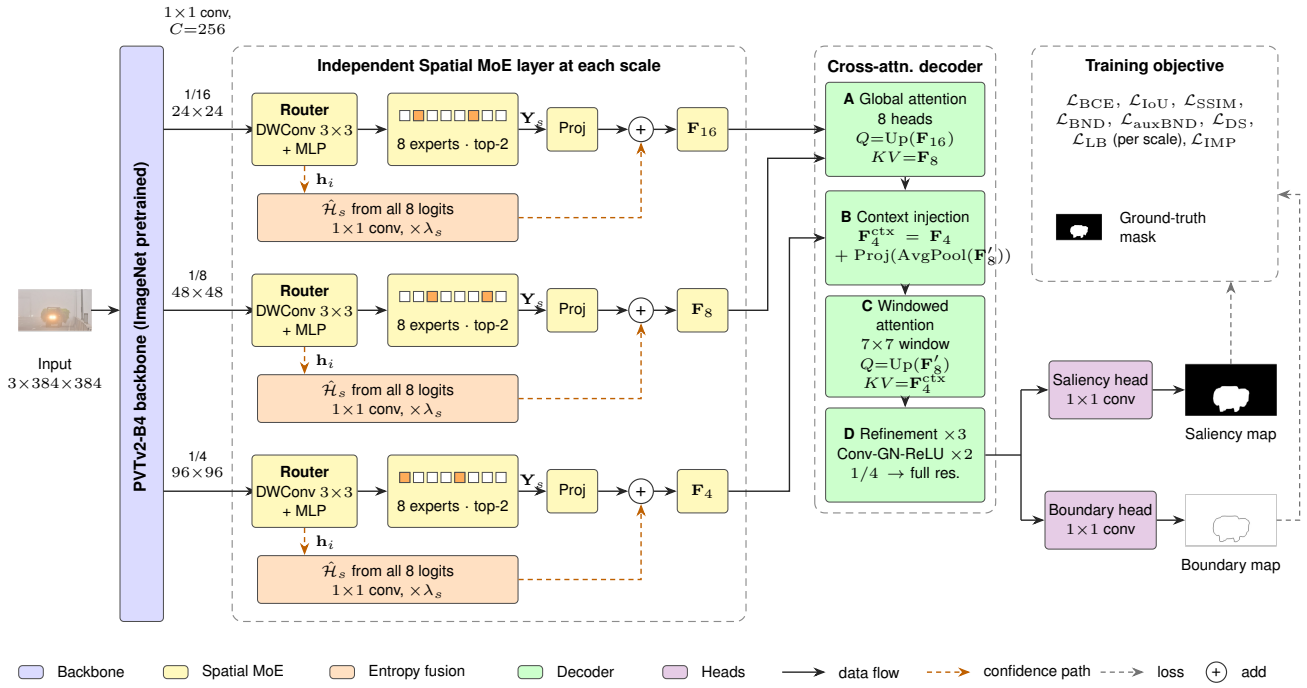


Fig. 2 SpatialMoE-SOD. A PVTv2-B4 backbone produces features at three resolutions. Each scale has its *own* Spatial MoE layer: a depthwise-convolutional router turns every token into eight logits, noisy top-K=2 sends the token to two of eight experts, and the experts' outputs are combined with their gate weights. Per-token routing entropy $\hat{\mathcal{H}}_s$, taken over the full eight-way distribution and normalised by log E, is projected and added to the routed features. The cross-attention decoder merges the three scales coarse-to-fine, and two 1×1 convolutions produce the saliency and boundary maps. Eight loss terms are active, including deep supervision over three auxiliary saliency heads at the 1/16, 1/8 and 1/4 scales.

3.1 Multi-Scale Features

A PVTv2-B4 backbone [25] pretrained on ImageNet extracts features at three scales via `out_indices`=(0, 1, 2). Each scale is projected to C=256 via 1×1 convolution:

$$\mathbf{X}_s = \text{Proj}_s(\text{Backbone}(\mathbf{X})_s) \in \mathbb{R}^{C\times H_s\times W_s}, \tag{1}$$

where $s \in \{4, 8, 16\}$ with $(H_s, W_s) \in \{(96, 96), (48, 48), (24, 24)\}$. Each feature map is reshaped into $N_s = H_s \times W_s$ tokens $\mathbf{x}_i \in \mathbb{R}^C$.

3.2 Router

For each scale, a depthwise 3×3 convolution captures local spatial context from the feature map; the full routing mechanism is shown in Fig. 3. The router input concatenates each token with its local context:

$$\mathbf{r}_i = [\mathbf{x}_i \mid \text{DWConv}_{3\times3}(\mathbf{X}_s)_i] \in \mathbb{R}^{2C}, \tag{2}$$

where | denotes channel-wise concatenation, yielding a $2C$-dimensional vector that encodes both the token's content and its spatial neighborhood. A two-layer MLP produces routing logits:

$$\mathbf{h}_i = W_2 \cdot \text{GELU}(W_1\mathbf{r}_i + b_1) + b_2 \in \mathbb{R}^E, \tag{3}$$

with $W_1 \in \mathbb{R}^{128\times2C}$, $W_2 \in \mathbb{R}^{E\times128}$, E=8 experts. The hidden dimension 128 provides a bottleneck between the $2C$=512 input and E=8 output.

3.3 Noisy Top-K Routing

During training, Gaussian noise is added with input-dependent scale to encourage exploration:

$$\tilde{h}_{i,j} = h_{i,j} + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma_j^2(\mathbf{x}_i)), \tag{4}$$

where $\sigma_j(\mathbf{x}_i) = \text{softplus}(w_j^\top \mathbf{x}_i + b_j)$ is a learned per-expert, per-token noise scale. At inference, noise is disabled (ϵ=0). Top-K=2 experts are selected per token; gate weights are softmax over selected logits only:

$$g_{i,k} = \frac{\exp(\tilde{h}_{i,k})}{\sum_{j\in\mathcal{T}_i}\exp(\tilde{h}_{i,j})}, \quad k \in \mathcal{T}_i = \underset{j}{\text{top-k}}(\tilde{h}_{i,j}, K{=}2). \tag{5}$$

This produces a sparse assignment: each token attends to exactly 2 of 8 experts, and each expert processes only the tokens that selected it.

3.4 Sparse Expert Aggregation

Each expert $\mathcal{E}_k$ executes only on tokens that selected it. For expert k, let $\mathcal{A}_k = \{i : k \in \mathcal{T}_i\}$ denote its assigned tokens:

$$\mathbf{y}_i = \sum_{k\in\mathcal{T}_i} g_{i,k} \cdot \mathcal{E}_k(\mathbf{x}_i), \tag{6}$$

implemented via a loop over experts with mask-based selection (true sparsity, not dense with masking). Each expert is a token-wise residual MLP:

$$\mathcal{E}_k(\mathbf{x}) = \mathbf{x} + W_k^{(2)} \cdot \text{GELU}(W_k^{(1)} \cdot \text{LN}(\mathbf{x})), \tag{7}$$

with $W_k^{(1)} \in \mathbb{R}^{4C\times C}$, $W_k^{(2)} \in \mathbb{R}^{C\times4C}$ (4× expansion from C=256 to $4C$=1024). Each scale has 8 independent experts

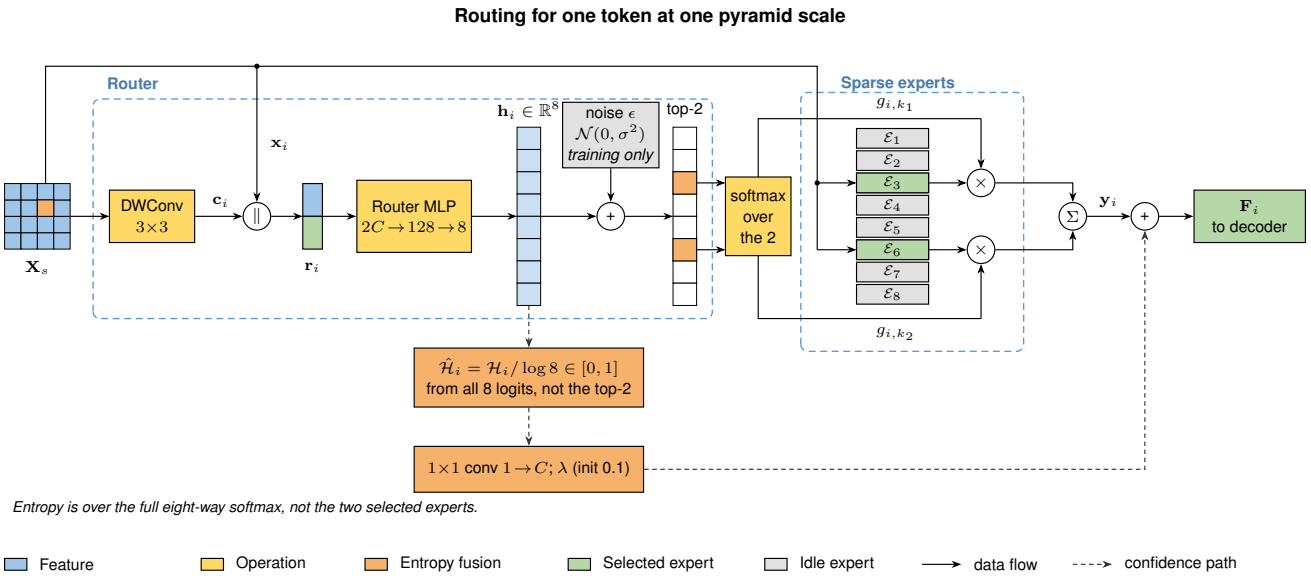


Fig. 3 Routing for one token at one pyramid scale. A depthwise 3×3 convolution supplies local context, which is concatenated with the token and passed through a two-layer MLP to give eight routing logits. Input-dependent Gaussian noise is added during training [9]; top-K=2 then selects two experts, and the gate weights are a softmax over those two logits only. Each selected expert is a residual MLP with a 4× channel expansion. Routing entropy is taken over the *full* eight-way distribution, not the selected pair, and the normalised value is projected and added to the decoder feature.

with no weight sharing (24 total). Reshaping per-token outputs yields $\mathbf{Y}_s \in \mathbb{R}^{C\times H_s\times W_s}$.

3.5 Routing Entropy

Per-token routing entropy measures assignment uncertainty over the full expert distribution:

$$\mathcal{H}_i = -\sum_{j=1}^{E} p_{i,j}\log p_{i,j}, \quad \hat{\mathcal{H}}_i = \mathcal{H}_i/\log E \in [0,1], \quad (8)$$

where $p_{i,j}$ is the router's softmax probability for expert j, computed over all E experts, not only the selected ones. The normalizing constant is hard-coded to $\log E$, matching the eight-expert model, so $\hat{\mathcal{H}}_i \in [0,1]$: 0 indicates a single expert dominates and 1 a uniform assignment over all eight experts. The normalized entropy map $\hat{\mathcal{H}}_s \in \mathbb{R}^{1\times H_s\times W_s}$ is fused with MoE output via a learned additive block:

$$\mathbf{F}_s = \mathrm{Proj}_Y(\mathbf{Y}_s) + \lambda_s \cdot \mathrm{Proj}_{\mathcal{H}}(\hat{\mathcal{H}}_s), \quad (9)$$

where $\mathrm{Proj}_{\mathcal{H}}$ is a 1×1 convolution from 1 to C channels and λ_s is a learnable scalar initialized to 0.1. $\hat{\mathcal{H}}_s$ is computed from the router logits and is *not* detached before fusion, so the decoder is a second, direct, differentiable task-gradient path to the router, independent of the expert-difference term in Eq. (6). It is weak when the gate is near-uniform, because $\partial\hat{\mathcal{H}}/\partial h \to 0$ there, but it is not zero, a point that matters for the k=1 arm in Section 5.3.

3.6 Decoder

The decoder merges coarse-to-fine features via top-down cross-attention [26]. At the coarsest scale, $\mathbf{F}_{16}$ is upsampled and used as queries with $\mathbf{F}_8$ as keys and values (8-head global attention):

$$\mathbf{F}'_8 = \mathrm{GlobalCrossAttn}(Q = \mathrm{Up}(\mathbf{F}_{16}),\ KV = \mathbf{F}_8). \quad (10)$$

Global context is then injected into the fine scale: $\mathbf{F}_4^{\mathrm{ctx}} = \mathbf{F}_4 + \mathrm{Proj}_g(\mathrm{AvgPool}(\mathbf{F}'_8))$. $\mathbf{F}'_8$ is upsampled and used as queries with $\mathbf{F}_4^{\mathrm{ctx}}$ as keys and values using 7×7 windowed attention [27] for efficiency:

$$\mathbf{F}'_4 = \mathrm{WinCrossAttn}(Q = \mathrm{Up}(\mathbf{F}'_8),\ KV = \mathbf{F}_4^{\mathrm{ctx}}). \quad (11)$$

Three refinement blocks (each: Conv 3×3 → GroupNorm [28] → ReLU → Conv 3×3 → GroupNorm → ReLU → residual addition) progressively upsample from 1/4 to full resolution. Two 1×1 convolution heads produce saliency logits $\hat{\mathbf{M}}_{\mathrm{sal}}$ and boundary logits $\hat{\mathbf{M}}_{\mathrm{bnd}}$ at 384×384. The decoder additionally exposes three auxiliary saliency heads at the 1/16, 1/8 and 1/4 scales (`aux_logits_16`, `aux_logits_8`, `aux_logits_4`), which the reported configuration enables through deep supervision.

3.7 Training Objective

The objective is a weighted sum of the eight terms that are active in the reported configuration (the Z-loss and the router-confidence term are disabled):

$$\begin{aligned}\mathcal{L} = {} & \lambda_{\mathrm{BCE}}\mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{IoU}}\mathcal{L}_{\mathrm{IoU}} + \lambda_{\mathrm{SSIM}}\mathcal{L}_{\mathrm{SSIM}} \\ & + \lambda_{\mathrm{BND}}\mathcal{L}_{\mathrm{BND}} + \lambda_{\mathrm{LB}}\mathcal{L}_{\mathrm{LB}} + \lambda_{\mathrm{IMP}}\mathcal{L}_{\mathrm{IMP}} \\ & + \lambda_{\mathrm{auxBND}}\mathcal{L}_{\mathrm{auxBND}} + \lambda_{\mathrm{DS}}\mathcal{L}_{\mathrm{DS}}, \end{aligned} \quad (12)$$

with λ_{BCE}=λ_{IoU}=λ_{SSIM}=λ_{BND}=1.0, λ_{LB}=λ_{IMP}=0.01, $\lambda_{\mathrm{auxBND}}$=0.5 and λ_{DS}=0.4 (λ_{LB} is applied at each of the three scales). These weights describe the *reported* model. The ablation-family arms of Section 4.4 share the architecture but not the router-loss configuration: they were re-run under the current code with per-scale load-balance weights 0.08/0.15/0.10, λ_{IMP}=0.05 and λ_{Z}=0.001. So this Method section does not describe the ablation model exactly, and the difference matters for the router: the

Z-loss penalises $\log \sum_j e^{h_j}$ and therefore pushes router logits towards zero, which drives the gate softmax towards uniform. That makes it a direct candidate cause of the flat routing entropy examined in Section 5. $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy with logits between predicted saliency logits and ground-truth mask. $\mathcal{L}_{\text{IoU}}$ is the soft IoU loss: $1 - \text{mean}_b \left[\frac{\sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon}{\sum_{hw} \sigma(\hat{M}_b) + \sum_{hw} M_b - \sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon} \right]$. $\mathcal{L}_{\text{SSIM}}$ is 1 minus the structural-similarity index [29] between the sigmoid prediction and the mask. $\mathcal{L}_{\text{BND}}$ is a smooth-L_1 term between the magnitude of the predicted spatial gradient and the ground-truth boundary map [30]. $\mathcal{L}_{\text{LB}}$ is the load-balancing loss [9]: $\sum_s E \sum_j f_j^{(s)} P_j^{(s)}$, summed over the three scales, where f_j is the fraction of all routing slots assigned to expert j, computed from hard assignments and detached, and P_j is the differentiable mean gate mass. $\mathcal{L}_{\text{IMP}}$ is the importance loss: $\frac{1}{|S|} \sum_s (\text{std}(\mathbf{P}^{(s)}) / \text{mean}(\mathbf{P}^{(s)}))^2$, where $\mathbf{P}^{(s)} \in \mathbb{R}^E$ denotes the per-expert mean gate-mass vector at scale s. $\mathcal{L}_{\text{auxBND}}$ is binary cross-entropy between the boundary head's logits and the boundary map. $\mathcal{L}_{\text{DS}}$ is the mean binary cross-entropy of the three per-scale auxiliary saliency heads [31]. Those three 1×1 convolutions are the only parameters deep supervision adds, accounting for the difference between the 69,212,349-parameter model that omits them and the reported 69,213,120-parameter model.

4 Experiments

4.1 Setup

We evaluate on WXSOD [12] (https://github.com/C-water/WXSOD), which provides 12,891 training images (scene-aware 80/20 split), 1,500 synthetic test images across 9 weather categories, and 554 real-world test images across 5 categories (fog, rain, snow, dark, light). Images are aspect-preserving resized to 384×384; augmentations include horizontal flip (p=0.5), color jitter (p=0.5), and ImageNet normalization. Training uses DDP on 2 GPUs, AMP (FP16), AdamW optimizer [32] (lr 10^{-4} for both newly introduced modules and the backbone, weight decay 10^{-4}), a WarmupCosine schedule [33], gradient clipping at 1.0 and effective batch size 32; the backbone is frozen during the first epoch and unfrozen from epoch 1 onward. Two schedules were used, and this matters for reading every comparison below. The reported model **R** was configured for a 50-epoch cosine with 1% warmup and early-stopped at 14 epochs, so its learning rate had *not* annealed when its numbers were taken. The ablation-family arms were configured for 8 epochs (14 for A1) with a 4% warmup, so they did anneal within their own, shorter schedules. Neither is a converged 50-epoch run, and **R**'s headline numbers are not reproduced by any ablation-family arm. Model selection for **R** is by validation MAE on the scene-disjoint validation split, so no test split enters it. **R** is evaluated single-pass, whereas the ablation-family arms use flip-average TTA; a flip-TTA evaluation of **R** also exists (MAE 0.0167) but is not the reported row, so the **R**-versus-**Ref** rows mix TTA settings. All evaluation numbers are scored in original image coordinates: predictions are reverse-geometried out of the 384×384 padded model frame before any metric is computed.

Table 1 Parameter and compute budget of the reported model at 384×384.

Component	Params (M)	Share
PVTv2-B4 backbone	51.18	73.9%
Cross-attention decoder	5.19	7.5%
SpatialMoE (3 scales)	$3 \times 4.28 = 12.84$	18.6%
Total	**69.21**	100%
MACs (G, per image)	277.9	—

Metrics. We report MAE and the structure measure S_ϕ [34], the enhanced-alignment measure E_ϕ^{adp} [35] and the maximum F-measure $F_\beta^{\max}$ [36], together with boundary MAE and boundary F1. MAE is the mean absolute difference [37] between the continuous prediction and the binary ground truth. For the threshold sweeps of Fig. 8, given a threshold t and the corresponding binary mask, we report precision $P(t)$, recall $R(t)$ and $F_\beta(t) = (1+\beta^2)PR/(\beta^2 P + R)$ with β^2=0.3 [38], computed per image and averaged over the split.

The model has 69.21M parameters and 277.9G MACs per 384×384 input image; the split by component is given in Table 1.

What we find. Four results, each backed by a table below.

- **The mixture-of-experts does not change the prediction.** Removing it entirely leaves real-test MAE at 0.0195, identical to the full model, and routing at one pyramid scale instead of three gives the same 0.0195 (Table 5).
- **Competitive without weather labels.** MAE 0.0192 on the 1,500 synthesized test images leads every method in the published benchmark, including the weather-supervised WFANet (0.0229); 0.0168 on the 554 real images is second of nineteen (Table 4, Fig. 4).
- **The router's gate distribution is nearly uniform.** Routing entropy sits at 99.2% or more of its $\log E$ ceiling in every functional arm except the single-expert arm A3 (97.7% at 1/8), so the decoder's entropy channel is nearly constant (Tables 9 and 10).
- **Training budget matters more than any single design choice.** The same recipe at 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, against a seed spread of ±0.0004 (Tables 5 and 12).

4.2 Main Results

Table 2 reports the headline metrics on both test splits, together with the remaining metrics of the reported model; the same six metrics for every ablation arm are given in the appendix (Table 3). The model achieves MAE of 0.0192 on synthetic data and 0.0168 on real-world data. Real-world MAE (0.0168) is slightly lower than synthetic MAE (0.0192). Neither number is attributable to the routing mechanism under study; the backbone and the decoder were not ablated (Section 5). However, this comparison is confounded by test-set composition: the synthetic test set includes compound weather conditions (rain+fog, rain+snow, snow+fog) that are absent from the real-world test set, which may increase synthetic-test error independently of any domain-shift effect.

One caveat on the two boundary numbers: they are com-

Table 2 Performance of the reported model on the WXSOD test splits (synthetic N=1500, real N=554). The two boundary rows are computed from the binarised saliency mask, not from the boundary head (see text).

Metric	Synthetic	Real
MAE ↓	0.0192	0.0168
S_ϕ ↑	0.9140	0.9151
E_ϕ^{adp} ↑	0.9591	0.9530
E_ϕ^{mean} ↑	0.9585	0.9551
E_ϕ^{max} ↑	0.9633	0.9608
F_β^{max} ↑	0.9016	0.8936
F_β^{mean} ↑	0.8886	0.8747
Boundary MAE ↓	0.4817	0.6334
Boundary F1 ↑	0.5472	0.3601

puted from the *saliency* prediction: the continuous map is binarised at 0.5 and its morphological boundary band is compared against the ground-truth boundary band at zero spatial tolerance, so they measure the agreement of the mask edge with a thin ground-truth band, not the accuracy of the separately trained boundary head, whose output is never scored. Boundary F1 is 0.3601 on real test under that convention. The boundary head's contribution is therefore unmeasured, and we do not read these numbers as evidence about it.

4.3 Comparison With Published Methods

WXSOD publishes results for 17 SOD methods and its own weather-supervised baseline WFANet on exactly these two test splits, under the same metric family [12]. Table 4 places our model alongside twelve of them, spanning the range of real-split MAE from the two weakest entries to the strongest, and Fig. 4 ranks all nineteen entries by real-split MAE. On the synthesized split our model attains the lowest MAE in that table, including WFANet, which is trained with explicit weather-class labels. On the real split it is second of nineteen, behind WFANet and ahead of every other method. The published numbers were produced by their authors under their own protocol; ours are measured here, so this is a like-for-like comparison in dataset, splits and metrics rather than a re-implementation, and it is labelled as reported. Our model is also larger than WFANet (69.21M against 50.87M parameters), which we state plainly rather than present as an efficiency claim.

4.4 Ablations

Table 5 reports every arm we trained; arm codes are defined in the caption. The middle block shares one training recipe (router noise enabled, effective batch 32, 8 epochs), and each row changes one field relative to the reference **Ref**, with two exceptions we state plainly: A3 changes E and k together, and the A2–A6 arms were run at E=4 or E=2 while **Ref** uses E=8, so expert count varies within the block and the E=4, k=2 reference arm was never run. Rows within the block are comparable only with those differences in mind.

Three findings stand out. First, **removing the mixture-of-experts entirely leaves performance unchanged**: the no-MoE arm scores 0.0195 on real test against 0.0195 for the full model, with S_ϕ agreeing to 0.0010. Second, **reducing routing from three pyramid scales to one changes nothing** (0.0195), and replacing the routed experts with a single dense expert per scale leaves the point estimate unchanged (0.0191) but is the one architectural arm the paired test does not certify as equivalent (Table 6). Third, the one arm that clearly separates is A3, the single-expert arm: it costs 0.0017 MAE (0.0212 against 0.0195), roughly four times the seed spread of ±0.0004 measured across seeds 42–44 (Table 12). A3 changes both E and k (Section 5.3), so it does not isolate top-k.

(The A2, A4 and A5 point estimates of 0.0002, 0.0001 and 0.0004 sit inside that band; the paired equivalence test of Table 6 certifies A2, A4 and A6 as equivalent but not A5, whose 0.0005 difference is not resolvable either way at this sample size.)

That floor is measured on the reference configuration, and the remaining arms are single-seed, so their deltas are compared against a floor estimated elsewhere. A paired bootstrap [56] over per-image errors, with a TOST equivalence test [57,58], is reported in Fig. 1 and Table 6; it tests against a margin of ±0.001 rather than against that seed floor, and the arms share the reference's seed, so it is not an independent comparison.

Training budget is the largest single effect we measured: the same recipe and seed trained for 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, larger than any individual architectural difference (the largest, A3, is 0.0017) though not larger than their sum, and confounded with the number of optimizer steps. We therefore report the architecture as characterisation, not as a demonstrated improvement.

Fig. 1 summarises the paired comparison of Table 6, and Fig. 5 places every arm's accuracy beside its routing entropy. Fig. 7 renders the no-mixture comparison as an image: the

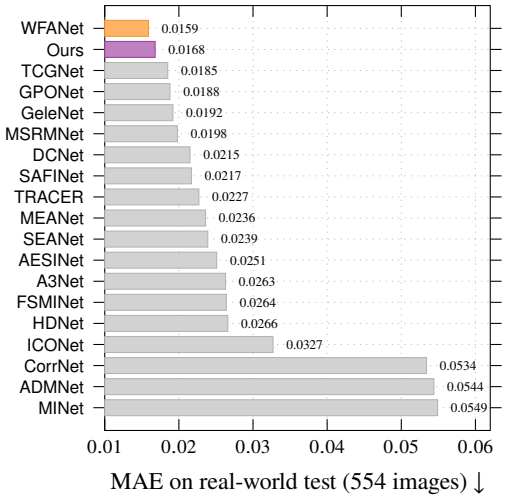


Fig. 4 Real-split accuracy ranking. All nineteen entries of the WXSOD benchmark tables, ordered by MAE. Grey: the 17 published SOD methods [39–55]; orange: WFANet, the benchmark's weather-supervised baseline; violet: ours. Every published value is as reported in [12] under the authors' own protocol; only the violet bar is measured in this work. Our model uses no weather labels at training or inference, whereas WFANet uses nine-class weather supervision. The horizontal axis starts at 0.010 rather than 0 so that the ranking is legible; it is not a zero-based scale.

Table 3 Extended metrics for every arm on the real (left) and synthetic (right) test splits. bMAE and bF1 are boundary MAE and boundary F1 (computed from the binarised saliency mask); arm codes follow Table 5.

	Real test (N=554)						Synthetic test (N=1500)					
Arm	E^{mean}	$E^{\max}$	F^{mean}	$F^{\max}$	bMAE	bF1	E^{mean}	$E^{\max}$	F^{mean}	$F^{\max}$	bMAE	bF1
R	0.9551	0.9608	0.8747	0.8936	0.6334	0.3601	0.9585	0.9633	0.8886	0.9016	0.4817	0.5472
Ref	0.9467	0.9567	0.8543	0.8854	0.7211	0.2876	0.9516	0.9617	0.8715	0.8925	0.5534	0.4947
A1	0.9554	0.9631	0.8719	0.8963	0.6675	0.3373	0.9536	0.9617	0.8812	0.8975	0.5067	0.5348
A2	0.9491	0.9583	0.8584	0.8860	0.7186	0.2912	0.9519	0.9611	0.8722	0.8922	0.5521	0.4968
A3	0.9376	0.9560	0.8293	0.8801	0.7688	0.2390	0.9440	0.9592	0.8514	0.8878	0.5874	0.4577
A4	0.9471	0.9571	0.8534	0.8860	0.7260	0.2835	0.9511	0.9604	0.8706	0.8917	0.5544	0.4941
A5	0.9492	0.9581	0.8583	0.8868	0.7174	0.2915	0.9512	0.9602	0.8715	0.8902	0.5540	0.4941
A6	0.9479	0.9581	0.8541	0.8841	0.7250	0.2835	0.9521	0.9615	0.8708	0.8909	0.5557	0.4907
A7	0.9460	0.9566	0.8525	0.8854	0.7317	0.2809	0.9508	0.9608	0.8692	0.8917	0.5612	0.4895
A8	0.9495	0.9592	0.8584	0.8869	0.7221	0.2881	0.9503	0.9588	0.8705	0.8889	0.5554	0.4929
A9	0.9452	0.9552	0.8498	0.8822	0.7345	0.2747	0.9514	0.9609	0.8688	0.8920	0.5559	0.4914
A10	0.9478	0.9583	0.8565	0.8854	0.7157	0.2922	0.9516	0.9604	0.8714	0.8911	0.5525	0.4945
B1	0.9469	0.9546	0.8535	0.8794	0.7345	0.2739	0.9516	0.9588	0.8687	0.8862	0.5676	0.4782
B2	0.9483	0.9559	0.8561	0.8804	0.7330	0.2768	0.9522	0.9592	0.8692	0.8862	0.5685	0.4771
B3	0.8181	0.8684	0.6375	0.7090	0.9314	0.0656	0.8427	0.8778	0.6714	0.7279	0.8539	0.1563

Table 4 Comparison with published WXSOD results. Published values are as reported in [12] (Tables II and III); ours are measured here. WFANet uses weather class labels, our model uses none. Lower MAE and higher S_ϕ are better.

	Synthetic		Real	
Method	MAE↓	S_ϕ ↑	MAE↓	S_ϕ ↑
MINet [39]	0.0775	0.7403	0.0549	0.7710
ADMNet [40]	0.0770	0.7497	0.0544	0.7839
TRACER [41]	0.0280	0.8898	0.0227	0.8961
SEANet [42]	0.0332	0.8694	0.0239	0.8978
A3Net [43]	0.0327	0.8784	0.0263	0.8909
DCNet [44]	0.0320	0.8792	0.0215	0.9058
MSRMNet [45]	0.0275	0.8895	0.0198	0.9091
SAFINet [46]	0.0384	0.8643	0.0217	0.9050
TCGNet [47]	0.0317	0.8832	0.0185	0.9185
GeleNet [48]	0.0239	0.9038	0.0192	0.9186
GPONet [49]	0.0266	0.8962	0.0188	0.9098
WFANet [12] *(weather-supervised)*	0.0229	0.9051	**0.0159**	**0.9248**
Ours *(no weather labels)*	**0.0192**	**0.9140**	0.0168	0.9151

full model and the arm with the mixture removed produce visually identical masks on the same scenes.

4.5 Inference-Time Ablations

Table 7 perturbs the *trained* reference model at inference only, so it isolates how much each component contributes to the final prediction rather than what training does. Replacing the learned router with **random routing** [59, 60] leaves real-test MAE within 0.0005 of the full model. Disabling entropy fusion changes it by 0.0001, and forcing every token at the 1/4 scale to a single expert changes it by −0.0001. None of the mechanism's parts affects the output: the mixture, its routing and its entropy channel are all inert at inference. Section 5 discusses why that may be so.

4.6 Weather-Wise Analysis

Table 8 reports per-condition performance on both test splits. On the real split, performance varies by 2.6× across weather types: snow yields the lowest MAE (0.0094, N=90), while light yields the highest (0.0243, N=93). Fog (0.0148, N=126), rain (0.0165, N=120) and dark (0.0191, N=125) fall between these extremes. The synthetic split spans nine conditions, three of them compound. The clean subset is the easiest (MAE 0.0142) and rain+fog the hardest (0.0237), which suggests that compositing degradations compounds difficulty; but the other two compound conditions (rain+snow, snow+fog) sit mid-table, below single-degradation light, rain and dark, so the ordering is not simply the number of degradation types.

Table 5 All trained arms. Real split N=554, synthetic N=1500. Arm codes: **R** the reported model (14 epochs, earlier recipe); **Ref** the ablation-family reference (8 epochs); **A1–A10** vary the reference: A1 14 epochs, A2 E=2, A3 E=4 k=1, A4 load-balance 0, A5 dense expert type, A6 no MoE, A7 dense gate, A8 routing at 1/4 only, A9 seed 43, A10 seed 44 (A2–A6 also change E from 8); **B1–B2** use an earlier, non-comparable recipe (batch 40), and **B3** is the divergent E=2, k=1 batch-40 run. TTA also differs across rows: **R** and B1/B2 are single-pass; A1–A10 and B3 use flip-average TTA.

	Real		Synthetic	
Arm	MAE↓	S_ϕ ↑	MAE↓	S_ϕ ↑
R	0.0168	0.9151	0.0192	0.9140
Ref	0.0195	0.9076	0.0214	0.9085
A1	**0.0174**	**0.9169**	**0.0202**	**0.9141**
A2	0.0193	0.9087	0.0210	0.9088
A3	0.0212	0.8985	0.0231	0.9036
A4	0.0196	0.9067	0.0212	0.9083
A5	0.0191	0.9086	0.0212	0.9080
A6	0.0195	0.9066	0.0210	0.9084
A7	0.0199	0.9060	0.0212	0.9084
A8	0.0195	0.9077	0.0212	0.9078
A9	0.0199	0.9048	0.0210	0.9086
A10	0.0191	0.9091	0.0213	0.9083
B1	0.0199	0.9034	0.0215	0.9041
B2	0.0197	0.9043	0.0215	0.9040
B3	0.0546	0.7714	0.0597	0.7827

4.7 Routing Entropy and Expert Usage

Table 9 reports the mean routing entropy per scale (left block). Entropy is taken over the full softmax over all E=8 experts, so the ceiling is $\log E = 2.0794$ nats; we therefore report both the raw value and its fraction of that maximum. The

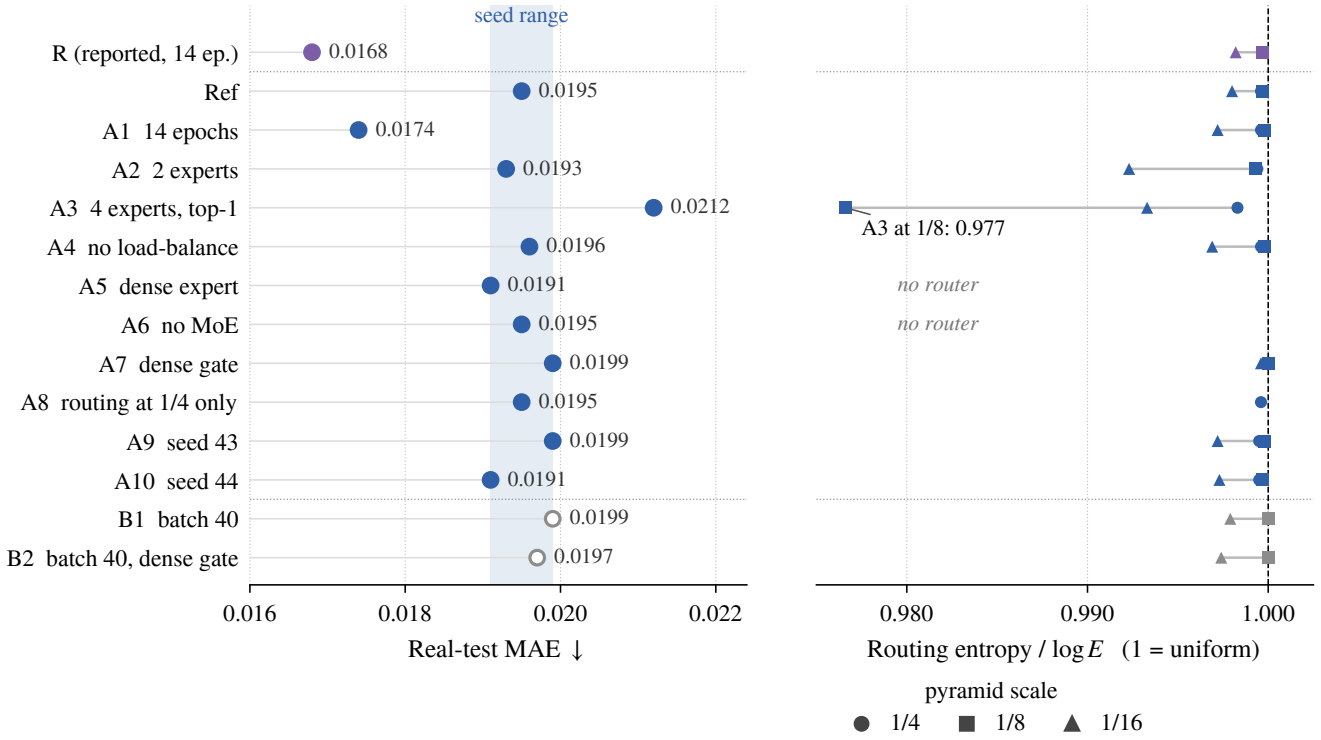


Fig. 5 **Accuracy and routing entropy for every trained arm.** Left: real-test MAE (Table 5); the shaded band is the range of the three seeds of the reference (Table 12). Filled markers are comparable one-recipe arms, hollow markers are the earlier batch-40 recipe, and R is the reported 14-epoch model. Right: routing entropy as a fraction of its log E ceiling at each pyramid scale (Table 10); arms with no router (A5, A6, and the non-routed scales of A8) have no entropy. Accuracy is flat across arms except for A3 and the longer schedule, and entropy stays at or above 0.992 of its ceiling in every routed arm except A3. B3, which diverged (MAE 0.0546), is off the axis.

Table 6 Each arm against the reference, on the 554 real-world test images. Δ is the paired mean difference in per-image MAE (negative = arm better); the interval is a 20,000-resample paired bootstrap over image indices; TOST tests $H_0\colon |\mu_\Delta| \geq 0.001$ against equivalence, and an arm is *equivalent* when $p < 0.05$. A2, A4, A6 and the routing-scope variant are equivalent to the reference within the margin; A5 (dense expert type) and the two seed replicates are not, and A3 and the 14-epoch run differ in the opposite direction. The architectural arms share the reference's seed, so their intervals are not independent. B2 and B3 are batch-40 arms (Section 4.4), not like-for-like with the S32 arms; A7 (dense gate) is characterised by its routing entropy (Table 10) rather than by a TOST here.

Arm	Δ MAE	95% CI	TOST p
A6 (E4 $k2$, MoE off)	−0.000019	[−0.000511, +0.000485]	0.0001
B2 (S40 E4 $k2$ gatedense)	+0.000075	[−0.000357, +0.000517]	0.0001
A8 (E8 $k2$, 1/4 only)	−0.000117	[−0.000562, +0.000354]	0.0001
A4 (E4 $k2$, no LB)	+0.000118	[−0.000282, +0.000506]	0.0000
A2 (E2 $k2$)	−0.000173	[−0.000473, +0.000124]	0.0000
Ref, seed 43	+0.000462	[−0.000153, +0.001172]	0.0566
A5 (E4 $k2$, dense)	−0.000488	[−0.001231, +0.000204]	0.0828
Ref, seed 44	−0.000497	[−0.001115, +0.000087]	0.0510
A3 (E4 $k1$)	+0.001599	[+0.001034, +0.002125]	0.9808
Ref, 14 epochs	−0.002224	[−0.002788, −0.001690]	1.0000
B3 (E2 $k1$ S40, diverged)	+0.035189	[+0.032444, +0.038059]	1.0000

measured mean sits within **0.2%** of the ceiling at every scale on both test splits: the gate distribution is nearly uniform in *magnitude*. That is a statement about magnitude, not about the *ranking* of the experts, and it is not a statement about indifference. The per-token hard assignment is in fact strongly skewed: at 1/8 three experts take about 84% of assignments and two are dead (Table 9, Fig. 10), so a near-uniform softmax coexists with a consistent, and possibly token-independent, ranking. Section 5.2 takes this up.

4.8 Expert Similarity and Usage

Table 9 compares expert pairs at each scale (right block). Pairwise cosine similarity between expert *weights* is essentially zero, so the experts are not copies of one another in parameter space. Cosine similarity between their *activations* on real data is 0.6424 at scale 1/4, 0.2733 at 1/8 and 0.8617 at 1/16. Those activation cosines are high because every expert adds the same residual $\mathbf{x}$ to its own branch; the quantity that would matter for re-routing is the cosine similarity of

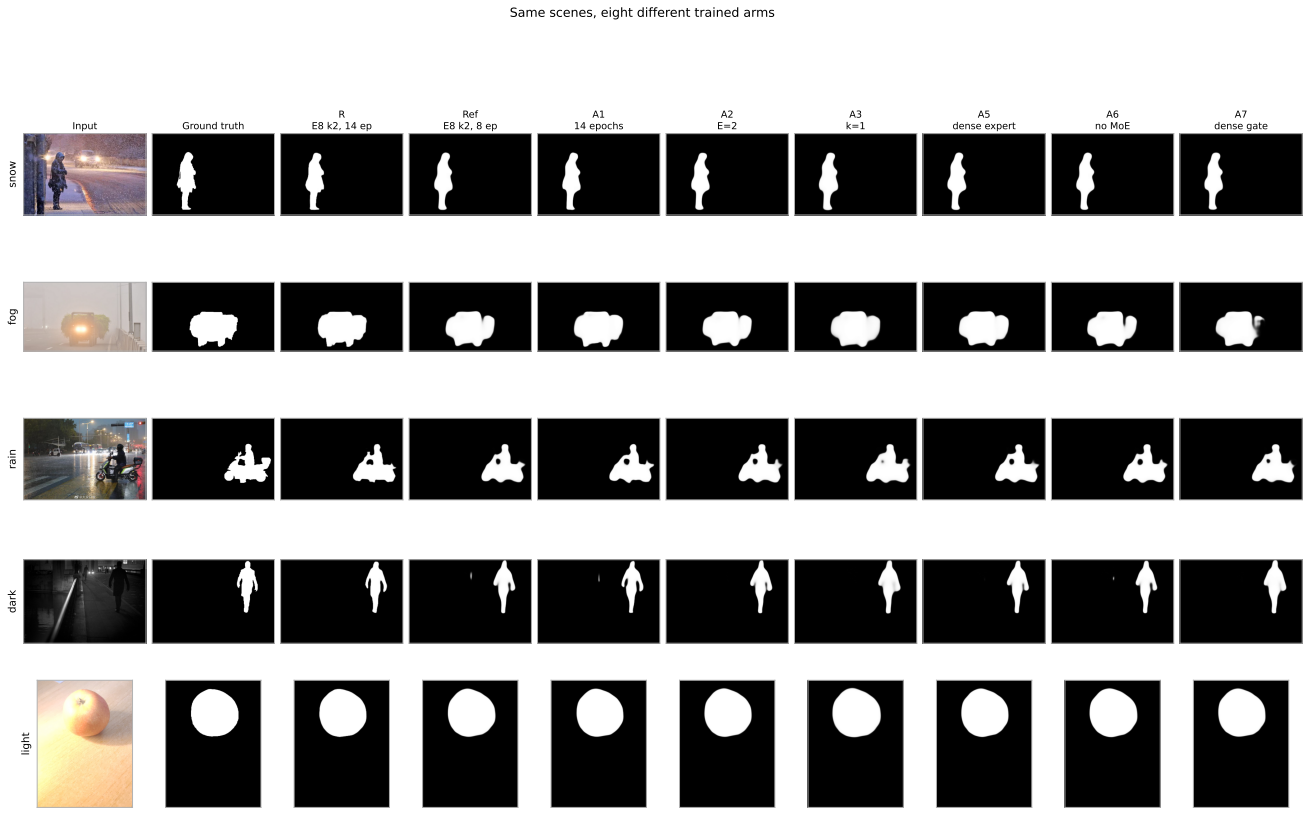


Fig. 6 **The same scenes predicted by eight different trained arms.** Rows are the five real-world test conditions, easiest to hardest by MAE: snow, fog, rain, dark and light. The first two columns are the input and the ground truth; the remaining eight are arms drawn from Table 5 spanning the widest architectural spread the study contains: the reported model, the ablation reference, a 14-epoch run, an E=2 model, an E=4 k=1 model, a single-dense-expert model, the no-mixture model, and the dense-gate model. Every one of those eight columns is a *different* model; the masks are nonetheless indistinguishable, and the two columns that do differ from the rest are the input and the ground truth. This is the ablation result of Table 5 rendered as an image, and it is the reason the paper reports a null result rather than a design win.

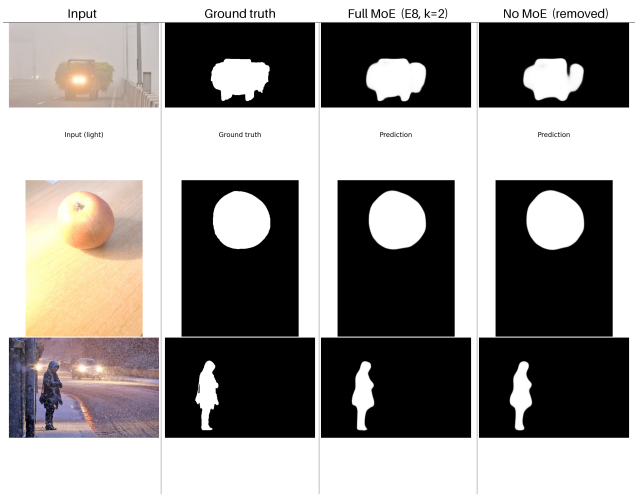


Fig. 7 **The mixture-of-experts does not change the prediction.** Read the four columns left to right: **(1) input**, **(2) ground truth**, **(3) full MoE** (E8, k=2, 14 epochs) and **(4) no MoE** (the identical architecture with the mixture removed). Columns 3 and 4 are therefore two different models on the same image; both are labelled "Prediction" inside the grid because that is how the evaluation tool renders a single model's output. Rows are real-world test images under fog, light and snow. The two prediction columns are indistinguishable in the light row; in the fog and snow rows the differences are confined to object boundaries. This is the qualitative counterpart of the no-MoE row in Table 5, where removing the MoE leaves real-test MAE unchanged at 0.0195.

Table 7 Inference-time ablations on the reference checkpoint (real test, N=554), re-scored in one pass. Random routing is the mean over 4 seeds. The full-model row therefore reads 0.0197 against the 0.0195 recorded for **Ref** in Table 5: the re-score is single-pass, where Ref's headline row uses flip-average TTA (worth about 0.0001 MAE). Compare rows within this table, not across the two.

Inference variant	MAE↓	S_ϕ ↑
Full model	0.0197	0.9043
Random routing (mean of 4)	0.0193	0.9060
Entropy fusion disabled	0.0197	0.9044
Force scale 1/4 to one expert	0.0196	0.9048

the *branch* differences, $\mathcal{E}_a(\mathbf{x}_i) - \mathcal{E}_b(\mathbf{x}_i)$, which we have not reported, and the weight similarity alone does not settle it. Tables 10 and 11 extend both reported measurements to every arm.

The gate's magnitude is near-uniform, but its ranking is not. The hard assignment is strongly skewed (Fig. 10), and the branch-difference similarity that would say whether the experts are actually distinguishable is not reported. Section 5 examines the candidates (noise-dominated selection at initialisation, auxiliary losses that do not reward decisiveness, and near-identical experts that leave a static selector) and states the diagnostic that would separate them.

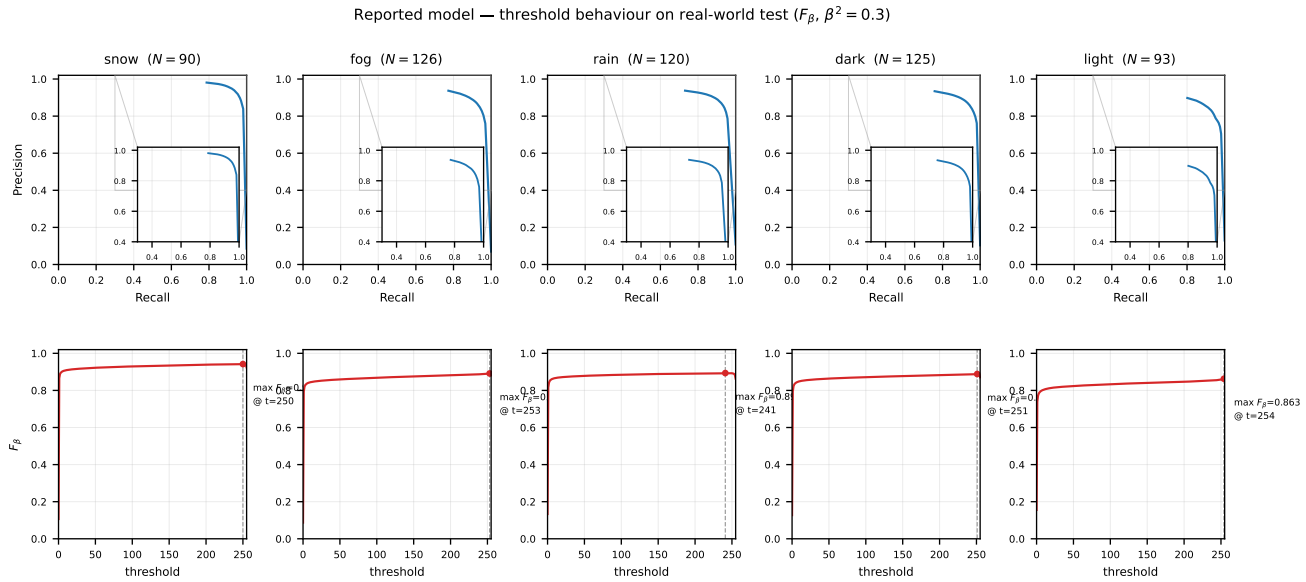


Fig. 8 **Precision–recall and F_β versus threshold, reported model.** One panel per real-world condition; the inset zooms the high-precision region. Curves are computed by sweeping the binarisation threshold over the continuous predictions and averaging $P(t)$, $R(t)$ and $F_\beta(t)$ (β^2=0.3) per image within each condition, then averaging over the condition. Only our model is plotted: no baseline predictions exist in this study, so this figure characterises the operating range of the reported model rather than comparing it with another method.

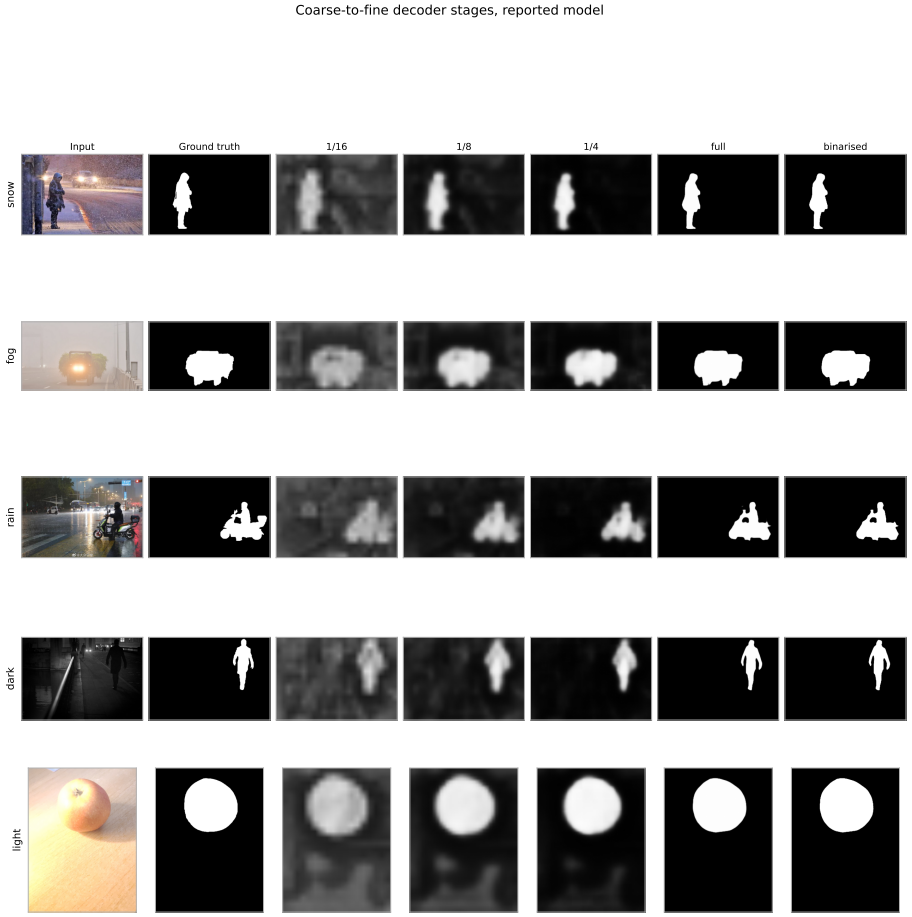


Fig. 9 **Progressive refinement through the decoder.** Each column is a successive stage of the top-down decoder, from coarse to fine: the saliency head applied to the $1/16$ stage features, then to the $1/8$ stage, then to the $1/4$ stage, then the full-resolution saliency map, and finally the binarised prediction beside the ground truth, on a representative scene per condition. The object is already located and largely outlined by the coarsest stage, and the later heads refine the boundary rather than change the decision.

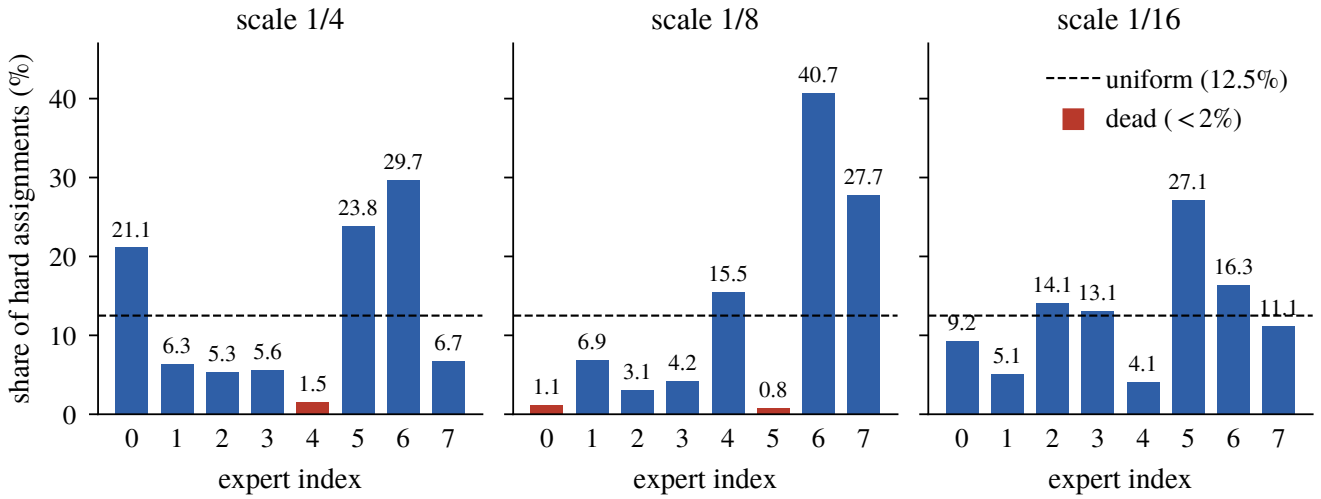


Fig. 10 **Hard-assignment expert usage per pyramid scale** on the reference arm at the final logged epoch. The dashed line marks uniform usage (12.5% for E=8); experts below 2% are dead (red). At 1/8 three experts take about 84% of assignments and two are dead, so a near-uniform gate softmax (Table 9) coexists with a strongly skewed ranking.

Table 8 Weather-wise results on the real (554 images) and synthetic (1,500 images) test splits, each ordered by MAE from best to worst. "—": not tabulated for the real split.

Condition	N	MAE↓	S_ϕ ↑	E_ϕ^{adp} ↑	$F_\beta^{\max}$ ↑
Real test					
Snow	90	0.0094	0.9478	—	0.9421
Fog	126	0.0148	0.9179	—	0.8914
Rain	120	0.0165	0.9157	—	0.8935
Dark	125	0.0191	0.9072	—	0.8885
Light	93	0.0243	0.8897	—	0.8629
Synthetic test					
Clean	167	0.0142	0.9329	0.9690	0.9229
Fog	166	0.0166	0.9308	0.9696	0.9255
Snow	172	0.0189	0.9192	0.9647	0.9068
Snow+fog	156	0.0189	0.9051	0.9567	0.8950
Rain+snow	166	0.0194	0.9152	0.9637	0.9032
Dark	156	0.0199	0.9049	0.9539	0.8836
Rain	179	0.0200	0.9173	0.9542	0.9030
Light	166	0.0215	0.9091	0.9623	0.9046
Rain+fog	172	0.0237	0.8903	0.9384	0.8691

4.9 Seed Variance

Table 12 repeats one configuration under three seeds. Real-test MAE spans 0.0008 (standard deviation 0.0003) and synthetic MAE spans 0.0004. This is the noise floor for every comparison in this paper: differences smaller than about 0.0005 MAE are not interpretable.

4.10 Discussion

The model demonstrates that spatial token-level routing is feasible for single-RGB SOD without weather-specific supervision. However, several important limitations must be noted. First and most centrally, the spatial MoE is shown across the eleven arms of the ablation family and three inference-time interventions to have no measurable effect on output quality; we treat that as the paper's primary contribution rather than a caveat. Routing entropy sits at or above 99.2% of its ceiling everywhere except the single-expert arm, so the decoder's entropy channel is nearly constant. Second, the comparison against prior methods uses the benchmark's published numbers rather than our own re-implementations, so it is a comparison as reported; WFANet still attains the best real-split MAE, so we do not claim state-of-the-art performance. Third, the reported boundary metrics are computed from the binarised saliency mask's edge against a thin ground-truth band at zero tolerance, not from the boundary head; F1 is 0.3601 on real test under that convention, and the boundary head's own accuracy is not measured here. Fourth, the model was trained once without replicate runs, so variance estimates for the reported configuration are unavailable. Fifth, the forced-expert analysis suggests that expert specialization may be limited, and the benefit of multi-expert routing over a single larger expert remains unclear.

5 Why the Router May Be Inert: Candidate Mechanisms

The ablation of Section 4.4 and the interventions of Section 4.5 establish *that* the router does not matter measurably, at this recipe. This section asks *why*. We keep the result and the explanation separate: the equivalence of the ablations is a finding, while the mechanism below is a ranked set of hypotheses, none of them established by a direct measurement of the router.

5.1 Routing as a Sparse Selection Operator

Let $\mathbf{X} \in \mathbb{R}^{N\times C}$ be the token matrix at one scale and $\mathcal{E}_j : \mathbb{R}^C \to \mathbb{R}^C$ the j-th expert. Write $\mathbf{\Pi} \in \{0,1\}^{N\times E}$ for the selection matrix induced by top-K, $\Pi_{ij} = \mathbb{1}[\, j \in \mathcal{T}_i \,]$, and let $\mathbf{g}$ be the $N \times E$ matrix of gate weights, supported on the selected experts. The layer is then exactly

$$\mathbf{y}_i \;=\; \sum_{j=1}^{E} (\Pi_{ij}\, g_{ij})\, \mathcal{E}_j(\mathbf{x}_i) \quad\Longleftrightarrow\quad \mathbf{Y} = (\mathbf{\Pi} \odot \mathbf{g}) \cdot \mathcal{E}(\mathbf{X}), \tag{13}$$

where $\odot$ is the Hadamard product and $\mathcal{E}(\mathbf{X}) \in \mathbb{R}^{N\times E\times C}$ stacks the expert outputs. The routing enters the output only

Table 9 Routing diagnostics per pyramid scale. *Entropy* (reported model, both test splits) is over the full distribution over all E=8 experts, so its ceiling is $\log E = 2.0794$ nats; values are shown in nats and as a percentage of that ceiling. *Reference arm* columns are measured at the final logged epoch: dead experts are those with hard-assignment usage below 2% (listed by index; per-expert usage is plotted in Fig. 10), and similarity is the mean pairwise cosine over expert pairs, on real test.

	Entropy (reported model)				Reference arm		
	Synthetic		Real		Dead	Cosine similarity	
Scale	nats	% of max	nats	% of max	experts	weights	activations
1/4	2.0786	99.96	2.0788	99.97	4	+0.0000	0.6424
1/8	2.0788	99.97	2.0789	99.97	0, 5	+0.0003	0.2733
1/16	2.0755	99.81	2.0756	99.82	none	+0.0010	0.8617

Table 10 Routing entropy per scale for every trained arm, normalised by $\log E$ (1.000 is a uniform distribution over the arm's expert pool), with the number of dead experts at each scale (hard usage below 2%). Arm codes follow Table 5. R is the reported checkpoint, measured by its own diagnostic pass; the rest are the ablation arms. "—" marks a scale with no router (hence no routing distribution and no dead-expert count): the two non-routed scales of A8, and all three scales of the dense-expert (A5) and no-MoE (A6) arms.

	1/4		1/8		1/16	
Arm	$H/\log E$	dead	$H/\log E$	dead	$H/\log E$	dead
R *(reported)*	0.9997	0	0.9997	2	0.9982	0
Ref	0.9996	1	0.9997	2	0.9980	0
A1	0.9996	1	0.9998	2	0.9972	0
A2	0.9994	0	0.9993	0	0.9923	0
A3	0.9983	1	0.9766	3	0.9933	3
A4	0.9996	0	0.9998	0	0.9969	0
A5	—	—	—	—	—	—
A6	—	—	—	—	—	—
A7	1.0000	0	1.0000	0	0.9996	0
A8	0.9996	1	—	—	—	—
A9	0.9995	1	0.9998	0	0.9972	1
A10	0.9995	1	0.9997	1	0.9973	0
B1	1.0000	0	1.0000	0	0.9979	0
B2	1.0000	0	1.0000	0	0.9974	0
B3	0.9993	0	0.9994	0	0.9958	0

Table 11 Mean pairwise cosine similarity between experts for every trained arm, over weight vectors and over activations, on real test. "—" means the quantity is undefined because the scale has fewer than two experts: the pass-through scales of A8, and the single shared expert of A5 and A6.

	1/4		1/8		1/16	
Arm	weights	activations	weights	activations	weights	activations
R *(reported)*	−0.0000	0.6383	+0.0002	0.2731	+0.0011	0.8748
Ref	+0.0000	0.6424	+0.0003	0.2733	+0.0010	0.8617
A1	+0.0000	0.6584	+0.0003	0.2621	+0.0011	0.8830
A2	−0.0002	0.7124	−0.0007	0.4914	+0.0007	0.8570
A3	+0.0007	0.5687	−0.0003	0.0739	+0.0011	0.8671
A4	+0.0007	0.6561	−0.0002	0.2998	+0.0010	0.8556
A5	—	—	—	—	—	—
A6	—	—	—	—	—	—
A7	+0.0000	0.7013	+0.0002	0.5607	+0.0015	0.9438
A8	+0.0000	0.6424	—	—	—	—
A9	−0.0005	0.6472	−0.0004	0.2631	+0.0002	0.8622
A10	+0.0001	0.6412	−0.0001	0.2790	+0.0006	0.8614
B1	−0.0002	0.6413	−0.0008	0.0795	+0.0008	0.8908
B2	+0.0006	0.6351	−0.0002	0.4469	+0.0011	0.8669
B3	−0.0004	0.6752	−0.0007	0.3939	+0.0009	0.7274

through the weighted selection $\mathbf{\Pi} \odot \mathbf{g}$; what that weighting acts on is the expert *difference*, which is where Section 5.5 begins.

5.2 The Router Is at Its Entropy Ceiling From Initialisation

The entropy of the gate distribution is the natural first quantity to inspect, but it cannot carry the argument on its own. Table 9 measures $\hat{\mathcal{H}} \in [0.9981, 0.9997]$ for the reported model, so

Table 12 Seed variance for E8 k=2, 8 epochs, everything else fixed.

	Real			Synthetic		
Seed	MAE	S_ϕ	E^{adp}	MAE	S_ϕ	E^{adp}
42	0.0195	0.9076	0.9402	0.0214	0.9085	0.9493
43	0.0199	0.9048	0.9380	0.0210	0.9086	0.9481
44	0.0191	0.9091	0.9412	0.0213	0.9083	0.9513

the gate distribution is within a few millibits of uniform *in magnitude*. Two caveats apply before any conclusion is drawn. First, magnitude is not ranking: top-K selection depends only on the *ranking* of the logits, and Fig. 10 shows that ranking is structured: at 1/8 three experts take about 84% of the hard assignments and two are dead. Second, any entropy functional has *zero* gradient at the uniform fixed point, so its inability to move the distribution is expected rather than informative. Entropy alone therefore establishes neither indifference nor failure to train.

Initialisation. Using the checkpoint's own configuration we build fresh models under five init seeds and read the normalised routing entropy at each scale from `MoEOutput.entropy`. At initialisation the mean over seeds is 0.9991, 0.9994 and 0.9993 of the $\log E$ ceiling at 1/16, 1/8 and 1/4. The router *begins* at its ceiling.

Training. The reported checkpoint's own diagnostic pass gives normalised entropies of 0.9997, 0.9997 and 0.9982 at 1/4, 1/8 and 1/16 (Table 10); a separate forensic pass over the same recipe on real-test tokens gives 0.9998, 0.9999 and 0.9968. Relative to initialisation the entropy rises slightly at 1/4 and 1/8 and falls at 1/16; the one scale that moved is the one that sharpened. What this shows is that the *magnitude* of the gate distribution barely changes. It does not show that the *top-2 assignment* is unchanged, because a router can reshuffle which experts it selects with almost no change in entropy; the measurement that would settle it (agreement between the initial and trained top-2 assignments on identical tokens) has not been made.

Consequence. If the router contributes approximately an unweighted mean of whichever experts it selects, deleting the mixture should cost nothing, and it does not: arm A6, in which the MoE is disabled outright, is statistically equivalent to the full model within ±0.001 MAE (Table 6; TOST $p = 0.0001$, paired bootstrap over the 554 test images), as are A2, A4

and the routing-scope variant, subject to the expert-count difference noted in Section 4.4. That equivalence is the paper's firm result. The explanation is not, and Section 5.5 ranks the candidates.

We describe the phenomenon as a router whose gate is near-uniform in magnitude but whose *ranking* is skewed and possibly token-independent (a static selector), rather than an indifferent or collapsed [61, 62] one. The hard usage at 1/8 (three experts, ≈ 84%; two dead) and the dead experts in nearly every arm support that reading; the branch-difference similarity that would confirm whether the experts are distinguishable is not reported.

5.3 Why the Single-Expert Arm Is the One That Separates

Most architectural variants in Table 6 are equivalent to the reference; the one that separates in the opposite direction is A3, which selects a single expert and moves MAE to 0.0212 against 0.0195. Two properties of that arm are structural rather than incidental.

First, with one selected expert the gate weight is not a softmax over a choice but the constant 1: renormalising over a single logit leaves 1 regardless of that logit's value. The layer therefore computes one expert's output rather than a mean of two, and the averaging that would suppress per-expert noise is absent.

Second, and this is why k=1 differs from every other knob, the gate receives no gradient through the *mixture*: with one selected expert the gate is the constant 1, so the sum over selected experts has a single term and no difference $\mathcal{E}_a - \mathcal{E}_b$ survives to carry a signal. That is not the same as receiving no task gradient at all. The router logits are still differentiable inputs to the decoder through the entropy channel (Section 3.5), which is not detached, and to the load-balance, importance and Z terms; the entropy path is weak for a near-uniform router because $\partial\hat{\mathcal{H}}/\partial h \to 0$ at uniformity, but it is not zero. The correct statement is that k=1 removes the mixture's own path to the router, not that the router is frozen.

We state the confound plainly: A3 was built at E=4 with k=1, while the reference and the other four-expert arms use k=2, so this arm varies E and k together and does not isolate k. The study also contains an E=2, k=1 run, but it diverged (0.0546; Table 5, B3) and is not evidence about k. Neither arm separates k from E or from a training failure; a matched E=2, k=2 versus E=2, k=1 pair at identical settings is what would isolate k, and we have not run one.

5.4 The Auxiliary Routing Losses Do Not Reward Decisiveness

Two of the auxiliary terms in Eq. (12) are *minimised by a uniform gate distribution*, and a third, active in the ablation recipe but not the reported one, penalises logit magnitude and therefore also pushes the gate towards uniform. They are a candidate explanation for the flat gate, though the ablation evidence below cuts against their being the main cause.

For the Shazeer load-balancing term, let f_j be the (detached) hard assignment fraction and P_j the differentiable mean gate mass, with $\sum_j f_j = \sum_j P_j = 1$. Because f is computed from the very assignments that produce P, the two are positively coupled, and under that coupling $\sum_j f_j P_j$ is minimised when both are uniform:

$$\sum_{j=1}^{E} f_j P_j \;\geq\; E \cdot \frac{1}{E} \cdot \frac{1}{E} \;=\; \frac{1}{E}, \tag{14}$$

with equality iff $f = P = u$. This is Chebyshev's inequality, so it requires f and P to be similarly ordered; they are, since both are computed from the same assignments, but the bound would fail for anti-correlated f and P. A concentrated routing table gives $\sum_j f_j P_j \to 1$, i.e. E times the minimum. The gradient of $\mathcal{L}_{\mathrm{LB}}$ with respect to the router parameters is $E \sum_j f_j \, \partial_\theta P_j$, which is non-zero for any non-uniform P and drives the gate mass *toward* uniform.

The importance term, $\mathcal{L}_{\mathrm{IMP}} = \big(\mathrm{std}(\mathbf{P})/\mathrm{mean}(\mathbf{P})\big)^2$, is likewise non-negative and exactly zero at uniform $\mathbf{P}$. Neither term rewards *decisiveness*; both penalise deviation from uniformity.

A third term is stronger still. The ablation-family recipe (Section 3) enables a Z-loss, $\mathcal{L}_{\mathrm{Z}} = \big(\log \sum_j e^{h_j}\big)^2$ with λ_{Z}=0.001, which the reported model disables. It is minimised by small router logits, and small logits make the softmax flat, so it pulls the gate towards uniform directly rather than merely failing to reward confidence.

The ablation evidence cuts against these terms being the main cause. Arm A4 removes the load-balance term (keeping the importance and Z terms at their family weights) and the router is *still* essentially uniform ($\hat{\mathcal{H}}$ = 0.9996, 0.9998, 0.9969 at 1/4, 1/8, 1/16), so removing load balance does not make the gate more decisive; and the reported model R has no Z-loss and is equally flat. A run with λ_{Z}=0 at the family recipe is the clean test of the Z-loss and has not been run.

Two limits apply, and an alternative design, expert-choice routing, removes the load-balance term entirely [63]. The load-balance and importance terms act on the batch-mean gate mass P_j, not on each token's own distribution, so a router could be balanced on average while every token is decisive, which is how such losses are normally used; per-token evidence would be needed to tie them to per-token uniformity. And none of these terms is shown here to *cause* the flat router: taken with the initialisation measurement, the defensible reading is that the objective contains no reward for decisiveness and at least one term that actively suppresses logit magnitude.

5.5 Ranked Hypotheses

Three mechanisms could leave the router near-uniform and unchanging. We state them with the evidence already in hand, which is suggestive but confounded by recipe differences, and the measurement that would discriminate each.

(H1) Selection is noise-dominated at initialisation. Noisy top-K adds Gaussian noise with a learned scale $\sigma_j(\mathbf{x}_i) = \mathrm{softplus}(w_j^\top \mathbf{x}_i + b_j)$, of order 0.7 at initialisation against logits of order 0.1; training-time selection would then be close to random [60]. *Evidence against:* the two noise-off batch-40 arms sit at 1.0000 at 1/4 and 1/8, flatter

than **Ref** with noise on, so removing the noise does not make the gate more structured. Confound: those arms also change the batch and gate mode.

(H2) The auxiliary and Z-loss terms do not reward decisiveness. The load-balance and importance terms are minimised at a uniform gate, and the ablation recipe's Z-loss penalises logit magnitude. *Evidence against their being the main cause:* arm A4 removes the load-balance term and the router stays equally flat, so load balance is not what makes it flat; and the reported model **R** has no Z-loss yet is equally flat, so the Z-loss is not the main cause either. They remain at most contributors.

(H3) The experts are near-identical at initialisation, leaving the gate nothing to separate. The mixture's own signal is $g_{ij}(\mathcal{E}_a - \mathcal{E}_b)$; with residual experts the shared $\mathbf{x}$ cancels, so the signal is the difference of two MLP branches, which is ≈ 0 if the experts start alike [64]. The gate is then a *static selector* [65,66]: small logits with a consistent, possibly token-independent ranking. Two observations support this and are not explained by gradient starvation [61]. First, the hard usage is strongly skewed with dead experts in nearly every arm, which a flat full-softmax entropy does not predict. Second, and opposite to a starvation story, arm A3, the arm with the least gradient through the mixture, is the *least* flat (0.9766 at $1/8$, with dead experts); this is one arm, confounded with expert count, so more starvation need not mean more flatness. Note that the dense-gate arm A7 being flat (Table 10) is *consistent* with H3, not evidence against it: a dense gate still multiplies the same expert difference. The discriminating measurements are the input-dependence of the assignment (decomposing the logit variance into token-independent and token-dependent parts, and the mutual information of the assignment with weather class and with spatial position), the initial-versus-trained top-2 agreement on identical tokens, and oracle routing by weather class as a positive control. None has been run.

None of the three is established; all three predict a near-uniform gate, and each is confounded. We report them as ranked hypotheses rather than a settled mechanism.

5.6 Summary: What Is Measured and What Is Hypothesised

Table 13 separates the two.

6 Limitations

The results in this paper are one training run per configuration, and the evidence for the null result rests on deltas that are small in absolute terms. We state the consequences.

The equivalence test is not an architectural comparison. Table 6 reports a paired bootstrap and a TOST against a margin of ±0.001 MAE chosen larger than the ±0.0004 seed floor and fixed after the measurements were in hand. The test fails to certify the two seed replicates ($p = 0.057$, $p = 0.051$) while certifying architecturally different arms that share the reference's seed 42, so it is measuring the shared per-image error structure at least as much as the architecture. Single-seed comparisons of this kind are known to be unreliable [67–69]; the clean benchmark is a seed-versus-seed comparison, and until each arm is trained under independent seeds, these equivalence claims are provisional [70].

The interventions cannot separate "useless" from "untrained". The inference-time perturbations (Table 7) replace or disable the router in a model trained with noisy routing; if selection was already noise-dominated during training (H1), then random routing at inference is in-distribution, and its small effect is expected. The interventions become informative only once a model has been trained with routing noise off.

The equivalence comparison does not use the reported recipe. **R** is the earlier-recipe model and was never re-run against a matched no-MoE control; its closest ablation-family counterpart, A1, differs from it by 0.0010 synthetic and 0.0006 real, larger than anything the paper treats as noise. A three-seed no-MoE versus full-model comparison at **R**'s recipe, and at 14 epochs, is what would turn single-run equivalence into a real comparison, and it has not been run.

The comparison is as reported. Every baseline number in Table 4 and Fig. 4 is the published authors' own measurement under their own protocol, not a re-run in this codebase. The one protocol difference we can state is the input convention (both score in original image coordinates). WFANet retains the best real-split MAE, so no state-of-the-art claim is made.

The decoder and the backbone were not ablated. All arms share a PVTv2-B4 backbone, a cross-attention decoder and the eight-term objective, so attributing performance to "the backbone" really attributes it to the backbone, decoder and loss together; nothing here separates them.

The E=4, k=2 reference arm was never run, so expert count is bracketed by E=2 and E=8 rather than measured at the midpoint.

One scale of the routing was probed. The forced-expert intervention was applied only to the $1/4$ scale; the effect of forcing the other two scales is unknown.

The boundary head is not measured. The reported boundary metrics come from the binarised saliency mask, not the boundary head (0.3601 F1 on real test under a zero-tolerance band overlap), so we cannot say whether the head is weak or how much it contributes.

The mechanism is a hypothesis. The measurements establish equivalence; they do not establish why. None of the three hypotheses of Section 5.5 has its discriminating measurement: neither the input-dependence of the routing nor the initial-versus-trained top-2 agreement has been computed, and none of the noise-off, small-σ, zero-Z-loss or oracle-routing runs has been performed.

Training-budget effects are unstudied. The 14-epoch versus 8-epoch comparison is a single pair of runs and confounds epoch count with the number of optimizer steps; we report it as the largest effect observed, not as an established scaling law.

7 Conclusion

We presented and ablated SpatialMoE-SOD. On WXSOD the model reaches MAE 0.0168 on the 554 real test images

Table 13 The firm measurements and the candidate mechanisms.

Statement	Status
The router begins at its entropy ceiling	**Measured**: mean 0.9991, 0.9994, 0.9993 of $\log E$ at initialisation over five seeds
Deleting the mixture is free	**Measured**: A6 equivalent to the reference, $\Delta = -0.000019$, TOST $p = 0.0001$ (Table 6)
(H1) Noise-dominated selection at init	**Not supported**: the noise-off arms B1/B2 are flatter, not sharper
(H2) Auxiliary / Z-loss terms	**Not the main cause**: A4 removes load balance and stays flat; R has no Z-loss and is equally flat
(H3) Near-identical experts / static selector	**Untested**: needs the input-dependence and initial-vs-trained top-2 agreement measurements; oracle routing is the positive control

and 0.0192 on the 1,500 synthetic ones. A complete ablation and inference-time intervention study shows the mixture-of-experts is not responsible for those numbers: it can be removed, reduced to one pyramid scale, or replaced with a single dense expert, and the no-MoE, one-scale and no-load-balance variants each sit within the seed-noise band of the reference and pass a TOST equivalence test against it at a margin of 0.001 MAE (the dense-expert variant does not), though the arms differ from the reference in expert count as well as the named field and share its seed. We do not attribute the performance to the backbone or the decoder either, because neither was ablated; the supportable claim is that it is not attributable to the mixture, at this one recipe. Fourteen epochs instead of 8 exceeds every architectural effect we measured.

The paper is deliberately two-sided. Empirically, it reports weather-label-free numbers together with the ablation showing that the mixture does not produce them. Analytically, it asks why the routing is inert and ranks three candidates: selection that is noise-dominated at initialisation; auxiliary and Z-loss terms that do not reward decisiveness; and experts that are near-identical at initialisation, leaving the gate nothing to separate and making it a static selector. We do not present the mechanism as settled: the measurements that would separate the three (the input-dependence of the routing, the initial-versus-trained top-2 agreement, and oracle routing as a positive control) have not been made. We claim nothing beyond this configuration.

Removing the mixture, reducing routing to one scale, replacing routed experts with a dense expert and removing the load-balancing loss all leave real-test MAE within the seed-noise band of the reference, while routing to a single expert costs 0.0017. The mixture is not earning its parameters on this task, and the useful direction is not more routing capacity but the training budget.

Future work should run the experiments that discriminate among the candidate mechanisms: noise off, a small initial σ, zero Z-loss, a router learning-rate multiplier, and oracle routing by weather class as a positive control. It should also add a multi-seed no-MoE versus full-model comparison at **R**'s recipe, and re-run the benchmark's baselines under a single unified protocol.

Data availability

All data used in this study are publicly available. We use the WXSOD benchmark of Chen et al. [12], available at https://github.com/C-water/WXSOD, comprising 12,891 synthetic training images (`train_sys`), 1,500 synthetic test images (`test_sys`) and 554 real-world test images (`test_real`). No new data were collected. The published comparison numbers of Table 4 and Fig. 4 are reproduced from that benchmark's reference tables and are identified as such wherever they appear.

Code availability

The implementation, experiment configurations and per-run evaluation artifacts are available at https://github.com/AviBapatu/Spatial-MoE-for-Adverse-Weather-Salient-Object-Detection [71]. The repository is *private at the time of writing* and will be made publicly available on completion of the project; the link above is the address at which it will appear. The routing-forensics outputs behind Section 5, including the paired-bootstrap and TOST results of Table 6 and the initial-versus-trained entropy measurement, are archived under `analysis_results/` in the same repository. The three figures produced by the companion analysis notebook (Figs. 6, 8 and 9) are regenerated by `moe-of-sod__paper_figures.ipynb` from the archived prediction maps and are distributed with the code.

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